

MANISH CHANDER

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EDUCATION

University of Texas at Austin – Austin, TX

May 2029

Bachelor of Science in Chemical Engineering; Minor in Semiconductor Science and Engineering

GPA: 3.9/4.0

Completed Courses: Mechanics, Electricity & Magnetism, Organic & Physical Chemistry, Differential Equations.

Spring '27 Courses: Advanced Packaging, Thermal Management for Microelectronics, Transport Phenomena, Applied Statistics.

EXPERIENCE

Nanoscale Design & Manufacturing Lab – Austin, TX

January 2026 – Present

Additive Manufacturing of Microsystems Project Intern

- Designing and manufacturing a custom digital light processing printer end to end — SolidWorks design, in-house machining, CMM flatness metrology, and closed-loop tip/tilt correction code.
- Formulated silver/acrylate photopolymer resins for additive fabrication of redistribution layers and interconnects used in advanced semiconductor packaging.
- Modeled photochemical kinetics and diffusion-driven transport, trained a PyTorch machine learning surrogate to predict resolution from exposure dose, development time, and composition, then inverted it to identify resolution-limiting mechanisms and recipes with lower defectivity and improved large-area uniformity.

Texas Materials Scientists and Engineers – Austin, TX

September 2025 – Present

Professional Development Chair (Aug 2026 – Present)

- Directing professional development initiatives, hosting CAD, FEA, welding, and soft-skills workshops for members while building out career and networking resources.

Thermal Hardware Engineer (Sep 2025 – Present)

- Designed and built a dual-zone thermal-management platform (two-plate system, independent Peltier modules, insulating barrier) achieving stable, well-separated temperature zones for controlled thermal testing.
- Integrated heat sinks and liquid cooling to improve heat transfer and temperature uniformity.

Longhorn Neurotech – Austin, TX

September 2025 – May 2026

EMG Mechanical Design Team Lead

- Led design of a compact EMG sensor enclosure integrating electronics, sensors, and a battery, soldering the assembly and routing power and signal pathways while maintaining electrical isolation and mechanical stability.

PROJECTS

Computational Materials Research – Austin, TX

January 2026 – May 2026

- Modeled proton transport and Grotthuss hopping mechanisms in battery materials using density functional theory (DFT) simulations in VASP, identifying performance-limiting transport pathways.
- Ran high-throughput computational screening across material compositions, comparing thermodynamic stability and electronic structure, to accelerate candidate identification.

CREDENTIALS

Texas Institute of Electronics

August 2026 – May 2027

Semiconductor Manufacturing Training

- Training spans CMOS process flow (oxidation, ion implantation, photolithography, etch, CMP, metallization), vacuum and gas delivery, and RF/plasma processing, plus statistical process control and Lean Six Sigma DMAIC applied to manufacturing quality analysis. Hands on chip manufacturing and clean room experience.

ADDITIONAL

- Honors: Biomedical Engineering Case Competition Finalist, Consulting Competition Semifinalist (hosted by Accenture), International Baccalaureate Diploma (May 2025).
- Technical Skills: *Process & Metrology* — CMOS process fundamentals, statistical process control (SPC), Lean Six Sigma (DMAIC), JMP, Clean Room, SEM, CMM; *Simulation & Modeling* — DFT (VASP), FEA/thermal (ANSYS); *Software & Design Tools* — Python (PyTorch), MATLAB, Onshape, SolidWorks, Fusion 360.